



Herd-level prevalence of contagious mastitis pathogens and insights into somatic cell count dynamics using bulk tank milk samples from dairy farms in Alberta, Canada

Waseem Shaukat,^{1,2,3} Diego B. Nobrega,^{1,2,3} Kristen N. Kalbfleisch,¹ John P. Kastelic,¹
David F. Kelton,⁴ and Herman W. Barkema^{1,2,3,5*}

¹Faculty of Veterinary Medicine, University of Calgary, Calgary, AB, T2N 4N1 Canada

²AMR–One Health Consortium, Calgary, AB, T2N 4N1 Canada

³WHO Collaborating Centre for Research on Antimicrobial Resistance and Appropriate Use of Antimicrobials, One Health at UCalgary, University of Calgary, Calgary, AB, T2N 2T8 Canada

⁴Department of Population Medicine, Ontario Veterinary College, University of Guelph, Guelph, ON, N1G 2W1 Canada

⁵Department of Community Health Sciences, Cumming School of Medicine, University of Calgary, Calgary, AB, T2N 5A1 Canada

ABSTRACT

Mastitis continues to be among the top 3 challenges for dairy production globally, resulting in substantial economic losses. Current pathogen-specific prevalence estimates are crucial for effective surveillance and disease control. The objective of this study was to estimate the true prevalence of herds positive for major contagious mastitis pathogens in Alberta, Canada, and to describe bulk tank SCC (BTSCC) patterns and dynamics. Using a serial cross-sectional study design, bulk tank milk (BTM) samples were collected from all active dairy producers in Alberta in December 2021 ($n = 484$), April 2022 ($n = 486$), and October 2022 ($n = 484$) and tested for *Staphylococcus aureus*, *Streptococcus agalactiae*, and *Mycoplasma bovis* using a commercial multiplex quantitative-PCR kit. The BTSCC data for 2021 and 2022 were obtained for each herd for each milk pick-up and production-weighted geometric mean BTSCC (PWgBTSCC) was calculated. True prevalence was estimated using Bayesian latent class models. Mixed-effect modified Poisson and linear regression models were used to assess associations among pathogen positivity, herd characteristics, and PWgBTSCC. True prevalence of *Staph. aureus*-positive herds were 17.8%, 11.0%, and 13.8% in December, April, and October, respectively, whereas for *Strep. agalactiae*, true prevalence was consistent at 0.6%. For *M. bovis*, true prevalence was 2.0%, 4.7%, and 3.6% in December, July, and October, respectively. The prevalence of *Staph. aureus*-positive herds was higher (prevalence ratio [PR] = 1.31) among medium-sized herds (3,600–7,200 L/d milk delivered) compared with smaller herds (<3,600 L/d milk

delivered). Hutterite colony herds were more frequently *Staph. aureus*-positive at least once (PR = 1.23) compared with non-Hutterite herds. Non-Hutterite colony herds in central and southern regions were less frequently *Staph. aureus*-positive compared with the north region; however, this trend was opposite among Hutterite colony herds. Annual PWgBTSCC for Alberta dairy herds from November 16, 2021, to November 15, 2022, was 154,000 cells/mL (median [first quartile–third quartile]: 118,000–202,000). Herds with a medium (100,000–200,000 cells/mL) or high (>200,000 cells/mL) annual PWgBTSCC were more frequently *Staph. aureus*-positive (PR = 1.58 and 2.40, respectively) compared with <100,000 cells/mL herds. Annual PWgBTSCC was elevated for herds testing positive for any major contagious mastitis pathogen at least once compared with consistently negative herds. Herds in the south region had an 11% lower annual PWgBTSCC compared with north-region herds. These findings provided updated prevalence estimates of major contagious mastitis pathogens and critical insights into BTSCC patterns across herd types and regions, supporting targeted within-herd diagnostic follow-up and evidence-based mastitis control strategies.

Key words: *Staphylococcus aureus*, *Streptococcus agalactiae*, *Mycoplasma bovis*, dairy farms, surveillance

INTRODUCTION

The dairy industry faces substantial economic and operational challenges due to infectious diseases. This not only leads to increased spending on veterinary care and treatments but also reduces milk output, milk quality, and productive life for cows and fertility, with adverse effects on international trade of livestock and dairy products (Gröhn et al., 2005; LeBlanc et al., 2006; Barkema et al., 2015; Aghamohammadi et al., 2018).

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*Corresponding author: barkema@ucalgary.ca

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

Mastitis remains among the top 3 challenges for dairy herds worldwide and is one of the costliest diseases in dairy production (Bradley, 2002; Hogeveen et al., 2019). Economic impact of mastitis was estimated at Can\$662 (US\$520) per milking cow per year for a typical Canadian dairy herd (Aghamohammadi et al., 2018). Contagious pathogens such as *Staphylococcus aureus*, *Streptococcus agalactiae*, and *Mycoplasma bovis* are present in the udders of infected cows and are most commonly transmitted to healthy cows via direct contact (e.g., in the milking parlor) or through contaminated milking machine clusters, hands, towels, and so on. *Staphylococcus aureus* is a gram-positive bacterium often causing clinical and subclinical mastitis (Wilson et al., 1997), whereas *Strep. agalactiae* is a gram-positive bacterium that typically causes subclinical mastitis (Keefe et al., 1997). *Mycoplasma* spp. are pleomorphic bacteria lacking a cell wall that can cause chronic clinical mastitis and high SCC. The most commonly identified *Mycoplasma* spp. in bovine mastitis is *M. bovis* (Bushnell, 1984; Maunsell et al., 2011). In addition to mastitis, *M. bovis* also causes arthritis, otitis media, and respiratory diseases in cattle (Maunsell et al., 2011).

Knowledge of the prevalence and distribution of mastitis pathogens is crucial for devising effective mastitis control strategies because prevention, control, and treatment strategies for clinical and subclinical mastitis vary greatly based on etiology. Major contagious mastitis pathogens that are isolated from bulk milk, *Staph. aureus*, *Strep. agalactiae*, and *M. bovis* reside mainly (*Staph. aureus* and *M. bovis*) or only (*Strep. agalactiae*) in the udder; therefore, the presence of these bacteria in bulk tank milk (BTM) may indicate ongoing IMI in the herd caused by these bacteria. All 3 bacteria are associated with increased SCC (Keefe et al., 1997; Wilson et al., 1997; Maunsell et al., 2011). At the population level, BTM testing offers an efficient approach for screening herds and describing the broader distribution of contagious mastitis pathogens. However, interpretation of BTM results for mastitis control in individual herds still requires integration with cow- or quarter-level diagnostics and herd-specific management information.

Prevalence of *Staph. aureus*, *Strep. agalactiae*, and *Mycoplasma* spp. in BTM was estimated to be 74.0%, 1.6%, and 1.9%, respectively, in Prince Edward Island (Olde Riekerink et al., 2006) and 84.6%, 7.7%, and 2.6% in Québec (Francoz et al., 2012). A Canadian national multiprovince study based on 904 BTM samples of 226 herds across 9 provinces in 2003–2005 reported an overall province-stratified prevalence of 74%, 4.4%, and 0% for *Staph. aureus*, *Strep. agalactiae*, and *Mycoplasma* spp. (Olde Riekerink et al., 2010). Prevalence of *Strep. agalactiae* in Alberta decreased from 21% in 1986 to 11% in 1992 and 6% in 1993 (Schoonderwoerd

et al., 1993). Another study assessing spatio-temporal variation in prevalence of mastitis pathogens in milk samples cultures from Ontario between 2008 and 2017 reported that the proportion of *Staph. aureus* culture-positive samples increased from 15% in 2008 to 22% in 2017, whereas the annual odds of increase in prevalence of *Staph. aureus* was 1.06 (Acharya et al., 2021). However, herd-level prevalence of contagious mastitis pathogens in western Canada has not recently been reported.

Bulk tank SCC (BTSCC) is widely considered a measure of milk quality, with a higher BTSCC indicative of inferior milk quality (Bobbo et al., 2017; Moradi et al., 2021). Although BTSCC is measured for every milk shipment in Canada, these herd-level data are confidentially managed by provincial dairy boards. Provincial summary statistics periodically released by these boards are typically arithmetic means, weighted or unweighted. Geometric means, however, better represent herd-level milk quality because they are less influenced by occasional spikes. The 2014–2015 Canadian National Dairy Study (NDS) was the first to report national and provincial production-weighted geometric mean BTSCC (PWgBTSCC) values, estimating a mean PWgBTSCC of 195,000 cells/mL for Alberta and 208,000 cells/mL nationally (Bauman et al., 2018). Since then, no updated, stratified, or detailed characterizations of BTSCC data have been published, leaving a gap in current understanding of milk quality trends and their variation across Alberta herds.

The objectives of this serial cross-sectional study were, therefore, to (1) estimate herd-level prevalence of *Staph. aureus*, *Strep. agalactiae*, and *M. bovis* in Alberta dairy herds; (2) identify spatial distribution patterns and assess associations between herd-level characteristics and BTM positivity for these pathogens; and (3) evaluate BTSCC trends across various herd characteristics and determine associations of BTSCC with contagious mastitis pathogens and herd-level factors.

MATERIALS AND METHODS

The study was approved by the University of Calgary Veterinary Sciences Animal Care Committee (protocol AC21-0070).

Study Population

The study population included all operational dairy farms in Alberta that were delivering milk during any of the 3 designated sampling rounds between December 2021 and October 2022. These farms were categorized into northern, central, and southern regions, as described by Shaukat et al. (2024).

Sample Collection

At 3 moments in a year (December 2021 [T1], April 2022 [T2], and October 2022 [T3]), a 40-mL BTM sample was retrieved by bulk milk graders as an extra sample during routine milk collection from all active dairy producers in Alberta. Samples were well mixed and carefully obtained following the procedure outlined in Alberta's Bulk Milk Grader Manual and Study Guide (Agriculture and Forestry, Government of Alberta, 2020). Sealed samples were shipped at 4°C to the laboratory at the Faculty of Veterinary Medicine of the University of Calgary and stored at 4°C before processing and subsequent storage at -20°C.

Real-Time qPCR for Mastitis Pathogens

Bacterial DNA from *Staph. aureus*, *Strep. agalactiae*, and *M. bovis* was detected from BTM samples using Mastitis 4E (M4E; DNA Diagnostic A/S, Risskov, Denmark), a commercial multiplex real-time quantitative-PCR (qPCR) kit. Frozen whole-milk aliquots were thawed and mixed, and a representative 0.5-mL sample was taken to extract DNA of targeted pathogens, according to the manufacturer's protocol (DNA Diagnostic, 2017). The qPCR mixture included 5 µL of purified DNA and 15 µL of qPCR Master Mix, with amplification performed using real-time qPCR CFX96 (Bio-Rad, USA). The PCR results were interpreted according to the manufacturer's recommendations, with a cycle threshold ≤ 37 considered positive.

Data Management

Information regarding each farm's delivered milk volume for each collection during the corresponding period, geographic location (north, central, or south), and farm classification (Hutterite colony herds vs. non-colony herds) was sourced from Alberta Milk (Shaukat et al., 2024). Hutterite colonies are communal farming communities of German origin that operate mixed agricultural enterprises, including dairy, in western Canada, producing approximately 21% of Alberta's milk (Shaukat et al., 2024). Dairy herds from these colonies were classified separately from independently owned dairy herds, given our previous study found notable differences in the prevalence of infectious diseases between colony and non-colony herds (Shaukat et al., 2024). Average milk delivered volume retrieved over a year from each herd converted to average per day was used to categorize each herd as small, medium, or large, as described elsewhere (Shaukat et al., 2024). Additionally, BTSCC data were obtained from Alberta Milk for each dairy farm for each milk collection during 2021 and 2022.

Statistical Analyses

Statistical analyses were performed using Stata/SE 17.0 for Mac (StataCorp, 2021, College Station, TX) and R version 4.2.2 (<https://www.r-project.org/>), with P -value < 0.05 considered significant. An overview of the analytical workflow, including data sources, preprocessing, data integration, BTSCC metric derivation, statistical modeling, and visualization, is presented in Figure 1. Data were initially entered into spreadsheets using Microsoft Excel for Mac (Microsoft Corporation, Redmond, WA) and subsequently imported into R or Stata for analysis. Descriptive statistics were used to summarize herd characteristics and study outcomes. Categorical variables were reported as frequencies and percentages, whereas continuous variables were summarized using means and medians, along with first and third quartiles (Q1–Q3). For each pathogen, apparent herd-level prevalence at each sampling point was determined as the proportion of herds testing positive for that pathogen out of the total tested herds at that time point. Overall apparent herd-level prevalence for each pathogen was defined as the proportion of herds that tested positive at least once across the 3 sampling occasions. A sensitivity (Se) analysis was performed by excluding herds with < 3 samples. Additionally, the proportion of herds that tested positive at all 3 time points was also calculated.

Estimation of True Prevalence. To estimate true herd-level prevalence of each pathogen at each time point to account for diagnostic misclassification, we implemented a Bayesian latent class model using Markov Chain Monte Carlo (MCMC) simulation in R via JAGS version 4.3.1 (Plummer, 2003) through the *R2JAGS* package version 0.8–9 (Su and Yajima, 2024). The model incorporated a binomial likelihood for the number of observed PCR-positive herds (T) out of total tested (n) at each time point, as follows (Equation 1):

$$T \sim \text{dBin}(\text{AP}, n), \quad [1]$$

assuming that apparent prevalence (AP) was a function of Se, specificity (Sp), and true prevalence (TP), as follows (Equation 2):

$$\text{AP} = (\text{Se} \times \text{TP}) + [(1 - \text{Sp})(1 - \text{TP})], \quad [2]$$

where TP of herds positive for the given contagious mastitis pathogen at a given time point is a latent unknown parameter. We first parameterized beta-distributed priors for Se, Sp, and TP using the *epi.betabuster* function in the *epiR* package version 2.0.76 (Stevenson et al., 2024) as follows (Equations 3, 4, and 5):

$$Se = \text{dBeta}(\alpha_{Se}, \beta_{Se}),$$

[3]

$$Sp = \text{dBeta}(\alpha_{Sp}, \beta_{Sp}),$$

[4]

$$TP = \text{dBeta}(\alpha_{TP}, \beta_{TP}),$$

[5]

where α and β are shape parameters of the β distribution, derived by specifying the mode and either the 2.5th or 97.5th percentile, based on prior knowledge. Due to the absence of confidence or credibility intervals for some parameters, we conducted a sensitivity analysis to assess the impacts of alternative prior assumptions on posterior estimates. This involved using priors with

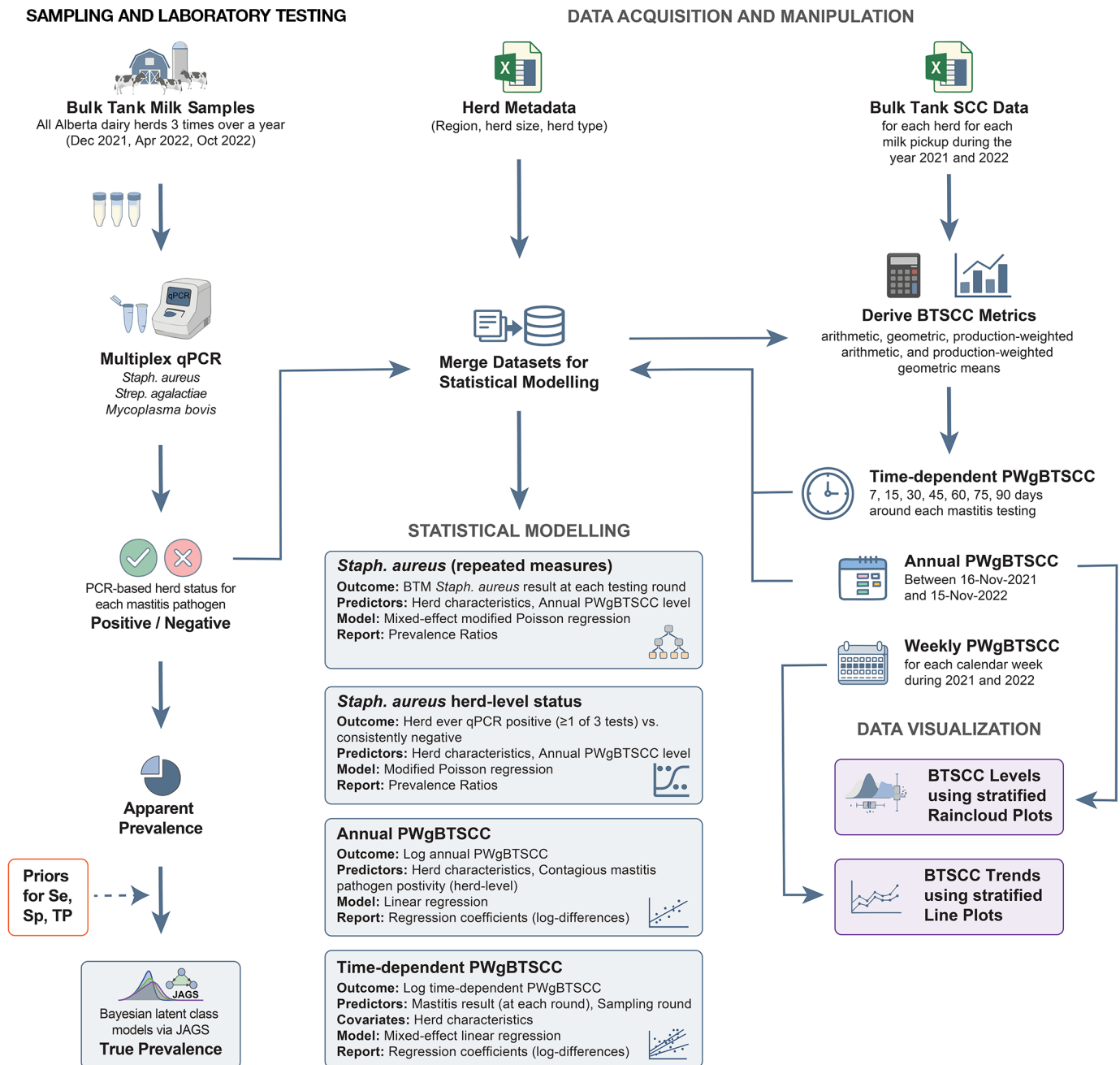


Figure 1. Graphical workflow summarizing data sources, preprocessing, data integration, bulk tank SCC (BTSCC) metric derivation, and statistical analyses used to estimate true prevalence, and evaluate associations between bulk tank milk (BTM) qPCR status for contagious mastitis pathogens, herd characteristics, and BTSCC metrics in Alberta dairy herds (2021–2022). PWgBTSCC = production-weighted geometric mean BTSCC.

Table 1. Priors for sensitivity, specificity, and prevalence, along with the resulting β distribution used as prior for the Bayesian latent class analysis to estimate true herd-level prevalence of contagious mastitis pathogens in Alberta dairy herds

Item	Prior estimates, % (95% CI)	Specified mode	97.5% confidence	Beta distribution (α , β)	Reference
<i>Staphylococcus aureus</i>					
Se	83.3 (68.1–92.1) ¹	0.833	>0.681	β (31.43, 7.09)	Soltau et al., 2017
Sp	66.7 (41.7–84.6) ¹	0.667	>0.417	β (11.32, 6.16)	Soltau et al., 2017
TP	22.0 (7.1–52.8) ²	0.220	<0.528	β (3.33, 9.27)	Bauman et al., 2018
<i>Streptococcus agalactiae</i>					
Se	92 (85.0–95.8)	0.920	>0.850	β (93.25, 9.02)	Zanardi et al., 2014
Sp	94 (88.0–97.1)	0.940	>0.880	β (107.06, 7.77)	Zanardi et al., 2014
TP	50 (10–90) ³	0.500	<0.900	β (2.09, 2.09)	Assumed
<i>Mycoplasma bovis</i>					
Se	15.3 (4.0–48.4)	0.153	<0.484	β (2.42, 8.84)	Nielsen et al., 2015
Sp	99.5 (98.9–99.8)	0.995	>0.989	β (1,148.9, 6.77)	Nielsen et al., 2015
TP	50 (2.5–97.5) ⁴	0.500	<0.975	β (1, 1)	Assumed

¹Se and Sp estimates taken from the referenced study, whereas 95% CI were calculated using data from the same study. The method explaining CI calculation is provided (Supplemental Table S2, see Notes).

²Prior true prevalence estimated using test positivity rates data from Bauman et al. (2018) using Se and Sp of qPCR and weak priors for prevalence with a mode of 50% with a 97.5% confidence that it is <90%.

³Vague priors.

⁴Flat or noninformative priors.

greater uncertainty around their modes and comparing the resulting model outputs. Additionally, as part of sensitivity analysis, we tested alternative priors for TP by modifying the mode or respective 2.5th or 97.5th percentile, or both. Priors for Se, Sp, and TP used in the final selected models—along with the specified mode (2.5th/97.5th) percentile and the resulting α and β shape parameters of each beta distribution—are presented in Table 1.

Four parallel and independent Markov chains, each initialized with different starting values, were run for 100,000 iterations, with a burn-in of 10,000 iterations and no thinning, except for *M. bovis*, for which 400,000 iterations per chain were run with 20,000 burn-ins to ensure adequate convergence. Model convergence was assessed using the potential scale reduction factor (Gelman–Rubin diagnostic) with an R-hat threshold <1.01, along with visual inspection of trace plots and autocorrelation plots generated using the *mcmcplots* package version 0.4.3 (Curtis et al., 2018). To assess the impacts of autocorrelation on the precision of parameter estimates, effective sample sizes (ESS) were calculated using the *effective-Size* function from the *coda* package version 0.19.4.1 (Plummer et al., 2006). A threshold of ESS $\geq 20,000$ was considered adequate to ensure robust characterization of posterior distributions; if ESS values were below this threshold, the number of iterations was increased incrementally until the criterion was met. Finally, model fit was evaluated using the deviance information criterion. Posterior distribution summaries and density plots were generated, and median and 95% Bayesian credible intervals (BCI) were reported as estimates of true prevalence for each pathogen at each time point.

Derivation of BTSCC Metrics. The BTSCC data were manipulated to generate various summary statistics for subsequent statistical analyses. First, annual (365 d) data were considered by filtering BTSCC records to include only samples collected from November 16, 2021, to November 15, 2022. For each herd, the following BTSCC metrics were calculated in R, using custom functions and the *dplyr* package for data manipulation (Wickham et al., 2014):

1. Arithmetic mean (aBTSCC): Calculated as the simple average of BTSCC values across all samples within the specified date range (Equation 6):

$$\text{aBTSCC} = \frac{1}{n} \sum_{i=1}^n x_i. \quad [6]$$

2. Production-weighted arithmetic mean (PWaBTSCC): Calculated by weighting each BTSCC value by the corresponding milk volume (Equation 7):

$$\text{PWaBTSCC} = \frac{\sum_{i=1}^n x_i w_i}{\sum_{i=1}^n w_i}, \quad [7]$$

where x_i is the BTSCC and w_i is the corresponding milk volume for observation i .

3. Geometric mean (gBTSCC): Calculated by log-transforming each BTSCC value, computing the

arithmetic mean of log-transformed values, and then exponentiating the result (Equation 8):

$$\text{gBTSCC} = \exp \left(\frac{1}{n} \sum_{i=1}^n \log(x_i) \right). \quad [8]$$

4. Production-weighted geometric mean (PWg-BTSCC): Calculated by multiplying each log-transformed BTSCC value by its corresponding milk volume, computing the weighted average of these values, and exponentiating the result using the natural exponential function (Equation 9):

$$\text{PWgBTSCC} = \exp \left(\frac{\sum_{i=1}^n \log(x_i) \cdot w_i}{\sum_{i=1}^n w_i} \right). \quad [9]$$

Subsequently, these 4 BTSCC summary metrics were calculated across multiple time windows of varying lengths (90, 75, 60, 45, 30, 15, and 7 d), each centered around the respective sampling time points (Supplemental Table S1, see Notes). In addition, all 4 metrics were computed for each calendar week from January 1, 2021, to December 31, 2022, for high-resolution temporal analysis. Annual PWgBTSCC was compared across herd characteristics using a Wilcoxon rank-sum test (for 2 groups) or Kruskal–Wallis test for >2 groups, followed by post hoc analysis using Dunn’s test with Bonferroni correction (Dinno, 2015).

Association of Mastitis Pathogen Positivity with Herd Characteristics and BTSCC. Prevalence ratios (PR) were used to assess associations between herd characteristics and BTM positivity for mastitis pathogens, as PR are more interpretable than odds ratios (OR) in cross-sectional studies with common outcomes, and are well suited to compare prevalences across strata of observed herd characteristics. Due to the limited number of herds testing positive for *Strep. agalactiae* and *M. bovis*, this analysis was restricted to *Staph. aureus*. We fitted mixed-effect generalized linear model with a log link and Poisson family in Stata, with herd included as a random effect to account for between-herd variability and within-herd correlation. The binary outcome variable was the *Staph. aureus* status of an individual BTM sample (negative vs. positive), whereas explanatory variables included herd size (small, medium, or large), herd type (Hutterite vs. non-Hutterite herds), geographical region (north, central or south), and annual PWgBTSCC level (low [$<100,000$ cells/mL], medium [$100,000$ – $200,000$ cells/mL], or high [$>200,000$ cells/mL]). Although Poisson regression is traditionally used for count data, it can be applied to binary

outcomes using a robust variance-covariance estimator (robust vce) to correct for model misspecification (Huber, 1967; White, 1980, 1982). This approach, referred to as modified Poisson regression, provides valid estimates of PR and their CI (Fonseca Martinez et al., 2017). Effect modification was assessed through 2-way interaction terms, using a backward elimination process, retaining only statistically significant interactions (Dohoo et al., 2010a,b). Model selection was guided by Akaike’s information criterion. Exponentiated coefficients from the final model (Equation 10) were interpreted as PR and reported with 95% CI and corresponding *P*-values:

$$\begin{aligned} \log(p_{(\text{SA})ij}) = & \beta_0 + \sum_{h=1}^2 \beta_{1(h)} \cdot (\text{herd size})_{ih} \\ & + \sum_{s=1}^2 \beta_{2(s)} \cdot (\text{PWgBTSSC}_{\text{year}})_{is} + \sum_{r=1}^2 \beta_{3(r)} \cdot (\text{region})_{ir} \\ & + \beta_4 \cdot (\text{herd type})_i + \sum_{n=1}^2 \beta_{5(n)} \cdot (\text{region} \times \text{herd type})_{in} + Z_{ui}, \end{aligned} \quad [10]$$

where $p_{(\text{SA})ij}$ is the expected probability of a PCR-positive BTM for *Staph. aureus* in the j th sample of the i th herd; β_0 is the intercept; β_1 through β_4 represent differences in log of the expected probability across categories of herd characteristics, relative to reference categories, holding other variables constant; β_5 is the change in the difference in logs of expected probability of *Staph. aureus* positivity between colony and non-colony herds with a change in geographical region; Z_{ui} is herd-specific random intercept; and h , s , and r represent categories of herd size, SCC level, and region, respectively.

Additionally, a generalized linear model with log link and Poisson family was fitted, where the outcome variable was the binary herd-level status for *Staph. aureus* (consistently negative herds vs. herds testing positive at least once), whereas explanatory variables were the same as previously described and the model was constructed using the previously described strategy to obtain the final model (Equation 11):

$$\begin{aligned} \log(p_{\text{SA}}) = & \beta_0 + \sum_{h=1}^2 \beta_{1(h)} \cdot (\text{herd size}) \\ & + \sum_{s=1}^2 \beta_{2(s)} \cdot (\text{PWgBTSCC}_{\text{year}}) \\ & + \sum_{r=1}^2 \beta_{3(r)} \cdot (\text{region}) + \beta_4 \cdot (\text{herd type}), \end{aligned} \quad [11]$$

where p_{SA} is the expected probability that a herd tested positive for *Staph. aureus* at least once during 3 time

points; and the coefficients β_1 through β_4 represent the log differences in expected probability across herd-level characteristics, relative to reference categories.

Association of Annual PWgBTSCC with Herd Characteristics and Mastitis Positivity. To determine the association of annual PWgBTSCC levels with herd characteristics and BTM-based mastitis status, a linear regression model was fitted in Stata, with log-transformed annual PWgBTSCC as the outcome. Explanatory variables included BTM-based mastitis status (herds consistently testing negative vs. herds testing positive at least once for any of the 3 pathogens), herd type, herd size, and geographical region. Model building followed the same approach described previously. Diagnostic plots supported assumptions of linearity, homoscedasticity, and normality of residuals. All variance inflation factors (VIF) were <2 , indicating minimal multicollinearity. Coefficients from the final model (Equation 12) were interpreted as log-transformed estimates relative to reference categories and reported with 95% CI and P -values:

$$\begin{aligned} \log(\text{PWgBTSCC}_{\text{year}}) = & \beta_0 + \beta_1 \cdot (\text{mastitis status}) \\ & + \beta_2 \cdot (\text{herd type}) + \sum_{h=1}^2 \beta_{3(h)} \cdot (\text{herd size})_h \\ & + \sum_{r=1}^2 \beta_{4(r)} \cdot (\text{region})_r + \varepsilon, \end{aligned} \quad [12]$$

where $\log(\text{PWgBTSCC}_{\text{year}})$ is the expected natural log of the annual PWgBTSCC (November 16, 2021, to November 15, 2022); β_0 is the intercept, representing the expected log-transformed PWgBTSCC for herds consistently negative for contagious mastitis pathogens, classified as small-sized, located in northern region, and designated as non-colony; β_1 is the difference in log-transformed PWgBTSCC between herds consistently negative and those testing positive at least once for any contagious mastitis pathogen; β_2 is the difference between colony and non-colony herds; β_3 and β_4 represent the difference in log PWgBTSCC for medium and large (relative to small) herds, and for central and southern (relative to northern) regions, respectively; and ε is the residual error term.

Association of Time-Dependent PWgBTSCC with Mastitis Positivity. To assess time-dependent impacts of mastitis positivity on PWgBTSCC, mixed-effects linear regression models were fitted in Stata using log-transformed PWgBTSCC values for multiple intervals (7, 15, 30, 45, 60, 75, and 90 d) surrounding each of the 3 mastitis test dates. Explanatory variables included BTM status for mastitis (negative for all vs. positive for any pathogen) and time point (T1, T2, and T3), with herd ID included as a random effect. Herd characteristics (region, herd size, and herd type) were assessed as potential

confounders. A covariate was considered a confounder if its removal from the model resulted in a meaningful change in the coefficient of the main effect (mastitis status). Effect modification was assessed using interaction terms; however, none were statistically significant. Model building followed a manual backward elimination strategy. Model diagnostics confirmed assumptions of linearity, homoscedasticity, and normality of residuals, with all VIF values <2 . Coefficients from final models were interpreted as log-transformed estimates and reported with 95% CI and P -values (Equation 13):

$$\begin{aligned} \log(\text{PWgBTSCC}_{ij}^{(d)}) = & \beta_0 + \beta_1 \cdot (\text{mastitis status})_{ij} \\ & + \sum_{t=1}^3 \beta_{2(t)} \cdot (\text{timepoint})_{ijt} + Zu_i + \varepsilon_{ij}, \end{aligned} \quad [13]$$

where $\log(\text{PWgBTSCC}_{ij}^{(d)})$ is the natural log-transformed PWgBTSCC for the d th time interval, corresponding to the j th observation in the i th herd; β_0 is the intercept, representing the expected log-transformed PWgBTSCC for herds with mastitis-negative BTM sample at T1; β_1 is the difference in log-transformed PWgBTSCC between mastitis-positive and mastitis-negative BTM samples when all other variables are held constant; $\beta_{2(t)}$ is the difference between t th time point compared with T1, when all other variables are held constant; Zu_i is herd-specific random intercept; and ε_{ij} is the residual error term for the j th observation in the i th herd.

Visualization. All data visualizations were performed in R using the *ggplot2* package, with multipanel figures constructed using *ggpubr* and *patchwork* packages. Raincloud plots were used to visualize the distribution of annual PWgBTSCC comparing herd subgroups defined by various characteristics and *Staph. aureus* status. Longitudinal trends in weekly PWgBTSCC over a 2-yr interval (January 1, 2021, to December 31, 2022) were visualized using time-series line plots, stratified by *Staph. aureus* status and herd characteristics to facilitate comparison of BTSCC patterns over time.

RESULTS

Herd Characteristics

During this study, 495 unique farms were included, of which 471 (95.2%) farms were tested at all 3 time points, 17 (3.4%) were tested twice, and 7 (1.4%) only once. These herds were distributed across 3 regions with 28%, 40%, and 32% located in the north, central, and south regions, respectively. Approximately 50% herds delivered $<3,600$ L/d milk and were categorized as small

herds, whereas 35% and 15% herds were classified as medium (3,600 to 7,200 L/d) or large herds (>7,200 L/d; Table 2). At the regional level, 58%, 44%, and 50% herds in the north, central, and south regions were categorized as small, whereas 11%, 23%, and 10% herds, respectively, were categorized as large. Approximately 13% herds had a PWgBTSCC_{year} of <100,000 cells/mL for the 1-yr interval, whereas the majority (61%) had a PWgBTSCC_{year} between 100,000 and 200,000 cells/mL (34%: 100,000–150,000; and 27%: 150,000–200,000), and 26% had a PWgBTSCC >300,000 cells/mL across Alberta. At the regional level, 13%, 11%, and 16% herds in the north, central, and south regions had PWgBTSCC_{year} <100,000 cells/mL, whereas 37%, 30%, and 13% herds, respectively, had a PWgBTSCC >300,000 cells/mL (Table 2).

Test Results

A total of 34.1%, 21.8%, and 27.1% herds tested positive for *Staph. aureus* at T1, T2, and T3, respectively, with 57.2% herds being PCR-positive at least once (Table 3). A total of 3, 4, and 4 herds tested positive for *Strep. agalactiae* at T1, T2, and T3, respectively; overall, 8 (1.6%) herds were PCR-positive at least once out of 3 time points (Table 3). Similarly, for *Mycoplasma bovis*, 1, 4, and 3 herds were PCR-positive at T1, T2, and T3, respectively, with an overall AP of 1.4% (Table 3).

Estimated True Prevalence

True herd-level prevalence for *Staph. aureus* varied across the 3 time points (Table 4). Posterior estimates of the true prevalence of *Staph. aureus*-positive herds was 17.8% (95% BCI = 5.55–32.9), 11.0% (95% BCI = 3.3–21.2), and 13.8% (95% BCI = 4.2–26.2) at T1, T2, and T3, respectively. For *Strep. agalactiae*, true prevalence remained consistent across time points, with posterior estimates of 0.6% (95% BCI = 0.1–1.8). For *Mycoplasma bovis*, true herd-level prevalence was estimated at 2.0% (95% BCI = 0.1–51.7), 4.7% (95% BCI = 0.2–60.3), and 3.6% (95% BCI = 0.1–56.1) at T1, T2, and T3, respectively (Table 4).

Association of *Staph. aureus* Positivity with Herd Characteristics

The prevalence of *Staph. aureus*-positive herds was higher (PR = 1.31 [1.08–1.58]; $P = 0.006$) among medium herds (3,600–7,200 L/d milk delivered) compared with smaller herds (<3,600 L/d). Herds with annual PWgBTSCC 100,000–200,000 cells/mL (PR = 1.58 [1.11–2.25]) and >200,000 cells/mL (PR = 2.40 [1.67–3.46]) were more frequently PCR-positive compared with herds with <100,000 cells/mL. In non-Hutterite colony herds, central and southern regions of Alberta had a lower prevalence (PR = 0.78 and 0.77, respectively) of *Staph.*

Table 2. Alberta dairy herds positive at least once for *Staphylococcus aureus*, *Streptococcus agalactiae*, and *Mycoplasma bovis* out of 3 time points across various farm characteristics

Item	No. (%) of farms	No. (%) of farms positive at least once		
		<i>Staph. aureus</i>	<i>Strep. agalactiae</i>	<i>M. bovis</i>
Total	495	283 (57.2)	8 (1.6)	7 (1.4)
Region				
North Alberta	139 (28)	84 (60.4)	2 (1.4)	1 (0.7)
Central Alberta	196 (40)	109 (55.6)	5 (2.6)	4 (2.0)
South Alberta	160 (32)	90 (56.3)	1 (0.6)	2 (1.3)
Herd size ^{1,2}				
Small (<3,600 L/d)	246 (50)	130 (52.9)	3 (1.2)	3 (1.2)
Medium (3,600–7,200 L/d)	172 (35)	106 (61.6)	1 (0.6)	3 (1.7)
Large (>7,200 L/d)	77 (15)	47 (61.0)	4 (5.2)	1 (1.3)
Farm type				
Hutterite colony	134 (27)	83 (61.9)	0 (0.0)	1 (0.8)
Non-colony	361 (73)	200 (55.4)	8 (2.2)	6 (1.7)
Bulk tank SCC ³ (cells/mL)				
<100,000	66 (13)	26 (39.4)	0 (0)	1 (1.5)
100,000–200,000	299 (60)	162 (54.2)	3 (1.0)	2 (0.7)
>200,000	130 (26)	95 (73.1)	5 (3.9)	4 (3.1)

¹Herd size is categorized based on delivered milk volumes with ≤3,600 L/d milk delivered as small, 3,600–7,200 L/d milk delivered as medium, and >7,200 L/d milk delivered as large.

²Approximate adult cows number estimated to ease interpretation of size categories using annual milk delivered by each farm and Alberta Milk's official statistics on average herd size and total Alberta milk production. This corresponds to approximately ≤125 adult cows for small, 125–250 adult cows for medium, and >250 adult cows for large herd size (Shaukat et al., 2024).

³Annual production-weighted geometric mean bulk tank SCC (November 16, 2021–November 15, 2022).

Table 3. Apparent prevalence (% and 95% CI) of *Staphylococcus aureus*, *Streptococcus agalactiae*, and *Mycoplasma bovis* in bulk tank milk of Alberta, Canada, dairy farms at 3 time points

Item	December 2021 (n = 484)	April 2022 (n = 486)	October 2022 (n = 484)	Overall prevalence ¹ (n = 495)	Positive 3 times (n = 471)
<i>Staph. aureus</i>					
North region	32 (24–40) ^a	27 (21–35) ^a	35 (27–43) ^a	60 (52–68) ^a	5 (2–10) ^a
Central region	33 (27–40) ^a	21 (16–27) ^{ab}	26 (20–33) ^{ab}	56 (49–62) ^a	3 (1–6) ^a
South region	37 (30–45) ^a	18 (13–24) ^b	21 (16–29) ^b	56 (48–64) ^a	2 (1–6) ^a
Alberta	34 (30–38)	22 (18–26)	27 (23–31)	57 (53–62)	3 (2–5)
<i>Strep. agalactiae</i>					
North region	0.8 (0.1–4.1) ^a	1.5 (0.4–5.2) ^a	0.7 (0.1–4.1) ^a	1.4 (0.4–5.1) ^a	0.8 (0.1–4.3) ^a
Central region	0.5 (0.1–2.9) ^a	1 (0.3–3.7) ^a	1.6 (0.5–4.5) ^a	2.6 (1.1–5.8) ^a	0.0 (0.0–2.0) ^a
South region	0.6 (0.1–3.5) ^a	0 (0.0–2.4) ^a	0.0 (0.0–2.4) ^a	0.6 (0.1–3.5) ^a	0.0 (0.0–2.4) ^a
Alberta	0.6 (0.2–1.8)	0.8 (0.3–2.1)	0.8 (0.3–2.1)	1.6 (0.8–3.2)	0.2 (0.0–1.2)
<i>Mycoplasma bovis</i>					
North region	0 (0.0–2.8) ^a	0.7 (0.1–4.1) ^a	0 (0.0–2.8) ^a	0.7 (0.1–4.0) ^a	0.0 (0.0–2.9) ^a
Central region	0 (0.0–2.0) ^a	1.6 (0.5–4.5) ^a	1 (0.3–3.7) ^a	2.0 (0.8–5.1) ^a	0.0 (0.0–2.0) ^a
South region	0.6 (0.1–3.5) ^a	0 (0.0–2.4) ^a	0.6 (0.1–3.5) ^a	1.2 (0.3–4.4) ^a	0.0 (0.0–2.4) ^a
Alberta	0.2 (0.0–1.2)	0.8 (0.3–2.1)	0.6 (0.2–1.8)	1.4 (0.7–2.9)	0.0 (0.0–0.8)
Any of the 3 contagious mastitis pathogens					
North region	32 (24–40) ^a	28 (21–36) ^a	35 (27–43) ^a	60 (52–68) ^a	6 (3–11) ^a
Central region	33 (27–40) ^a	22 (17–28) ^{ab}	27 (21–33) ^{ab}	56 (49–62) ^a	3 (2–7) ^{ab}
South region	38 (31–46) ^a	18 (13–24) ^b	22 (16–29) ^b	57 (49–64) ^a	2 (1–6) ^{ab}
Alberta	34 (30–39)	22 (19–26)	27 (24–31)	57 (53–62)	3 (2–5)

^{a,b}Values in the same column for same pathogen with different superscripts significantly differ ($P < 0.05$).

¹Overall prevalence refers to the percentage of farms that tested positive at least once in 3 time point tests.

Table 4. True herd-level prevalence of *Staphylococcus aureus*, *Streptococcus agalactiae*, and *Mycoplasma bovis* in Alberta, Canada, dairy farms using bulk tank milk samples at 3 time points (2021–2022)¹

Item	Prior knowledge scenarios, %			Posterior estimate of true prevalence, %		
	TP	Se	Sp	December 2021	April 2022	October 2022
<i>Staph. aureus</i>						
S1	22 (7–53)	83 (68–92)	67 (42–85)	18 (6–33)	11 (3–21)	14 (4–26)
S2	50 (10–90)	83 (68–92)	67 (42–85)	19 (4–36)	8 (2–21)	13 (3–27)
S3	60 (30–84)	83 (68–92)	67 (42–85)	29 (15–43)	18 (9–28)	22 (12–34)
S4	22 (7–53)	83 (30–97)	67 (42–85)	21 (6–46)	16 (4–43)	18 (5–44)
S5	22 (7–53)	83 (68–92)	67 (25–92)	21 (6–38)	15 (5–26)	18 (6–31)
S6	22 (7–53)	83 (68–92)	100 (70–100)	33 (13–45)	22 (9–31)	27 (11–37)
S7	22 (7–53)	83 (68–92)	100 (90–100)	38 (29–48)	24 (16–32)	30 (22–39)
S8	22 (7–53)	70 (40–89)	100 (70–100)	35 (13–54)	25 (10–42)	30 (11–48)
S9	22 (7–53)	60 (40–77)	100 (70–100)	38 (13–58)	27 (10–44)	32 (12–50)
S10	50 (10–90)	50 (10–90)	50 (10–90)	50 (10–89)	50 (9–91)	50 (9–90)
<i>Strep. agalactiae</i>						
S1	50 (10–90)	92 (85–96)	94 (88–97.1)	0.6 (0.1–1.7)	0.6 (0.1–1.8)	0.6 (0.1–1.8)
S2	60 (35–80)	92 (85–96)	94 (88–97.1)	2.6 (1.4–4.4)	2.8 (1.4–4.6)	2.8 (1.4–4.7)
S3	5 (1–30)	92 (85–96)	94 (88–97.1)	0.4 (0–1.5)	0.5 (0.1–1.6)	0.5 (0.1–1.6)
S4	50 (10–90)	88 (50–97)	94 (88–97.1)	0.7 (0.1–2.5)	0.8 (0.1–2.7)	0.8 (0.1–2.7)
S5	50 (10–90)	92 (85–96)	94 (50–98.8)	0.8 (0.2–2.1)	0.9 (0.2–2.3)	0.9 (0.2–2.3)
<i>M. bovis</i>						
S1	50 (2.5–97.5)	15.3 (4–48.4)	99.5 (98.9–99.8)	2.0 (0.1–52.0)	4.7 (0.2–60.9)	3.6 (0.1–56.8)
S2	50 (10–90)	15.3 (4–48.4)	99.5 (98.9–99.8)	13.3 (1.0–76.4)	16.3 (1.9–73.9)	15.0 (1.5–74)
S3	2.5 (1.1–5.7)	15.3 (4–48.4)	99.5 (98.9–99.8)	2.3 (0.9–4.9)	2.6 (1–5.2)	2.5 (1.0–5.1)
S4	50 (2.5–97.5)	15.3 (2.8–65)	99.5 (98.9–99.8)	3.3 (0.1–85.7)	6.1 (0.2–81.5)	5 (0.2–81.7)
S5	50 (2.5–97.5)	15.3 (4–48.4)	99.5 (80–99.8)	2.3 (0.1–70.1)	5.2 (0.2–85)	4.2 (0.2–84.9)
S7	17 (6.2–40)	15.3 (4–48.4)	99.5 (98.9–99.8)	11.3 (2.7–30.7)	12.2 (3.4–31)	11.9 (3.1–30.9)
S8	5 (1–30)	15.3 (4–48.4)	99.5 (98.9–99.8)	2.7 (0.3–14.8)	4.3 (0.6–17.6)	3.7 (0.4–16.7)
S9	10 (3.1–30)	15.3 (4–48.4)	99.5 (98.9–99.8)	6.1 (1.2–20.3)	7.3 (1.7–21.4)	6.9 (1.5–21)
S10	1 (0.6–49.9)	15.3 (4–48.4)	99.5 (98.9–99.8)	1.6 (0.1–16.3)	3.5 (0.2–22.2)	2.7 (0.1–20.1)

¹Median and 95% Bayesian credible interval estimates from the posterior distribution with the first row of each pathogen considered as the final model and its estimates, whereas subsequent rows are part of sensitivity analysis showing results with a unique set of alternate priors.

aureus–positive herds compared with the northern region. However, this trend was reversed in the Hutterite colony stratum, where central (PR = 1.70; $P = 0.047$) and southern (PR = 1.60; $P = 0.062$) region herds were more frequently PCR-positive than those in the north. Prevalence of *Staph. aureus*–positive herds was lower among Hutterite colony herds compared with non-colony herds in the north (PR = 0.61; $P = 0.046$), but this trend was opposite in central (PR = 1.33; $P = 0.073$) and southern (PR = 1.26; $P = 0.146$) regions (Table 5). Prevalence of herds PCR-positive at least once for *Staph. aureus* was high (PR = 1.23; $P = 0.020$) in Hutterite colony herds compared with non-colony herds (Table 5).

Bulk Tank SCC and Its Associations with Herd Characteristics and Mastitis Positivity

From November 16, 2021, to November 15, 2022, the annual production-weighted geometric mean BTSCC for Alberta dairy herds had a median of 154,000 cells/mL (Q1–Q3: 118,000–202,000) and a mean of 163,000 cells/mL (95% CI: 158,000–169,000; Table 6). The production-weighted arithmetic mean BTSCC for the same period had a median of 159,000 cells/mL (124,000–210,000) and a mean of 169,000 cells/mL (164,000–175,000; Table 7). Herds consistently testing negative for *Staph. aureus* had lower PWgBTSCC (median: 142,000 cells/mL [Q1–Q3:

Table 5. Prevalence ratios (PR) and 95% CI for farm characteristics and production-weighted geometric mean bulk tank SCC (PWgBTSCC) associated with *Staphylococcus aureus* detection in bulk tank milk samples using qPCR¹

Variable	Herds, no. (%)	Samples, no. (%)	Positive bulk tank test for <i>Staph. aureus</i>		Positive for <i>Staph. aureus</i> at least once	
			PR (95% CI)	<i>P</i> -value	PR (95% CI)	<i>P</i> -value
Herd size ^{2,3}						
Small	246 (50)	716 (49)	Referent		Referent	
Medium	172 (35)	508 (35)	1.31 (1.08–1.58)	0.006	1.22 (1.04–1.44)	0.017
Large	77 (15)	230 (16)	1.15 (0.89–1.48)	0.300	1.16 (0.94–1.44)	0.173
Annual PWgBTSCC ⁴						
<100,000 cell/mL	66 (13)	186 (13)	Referent		Referent	
100,000–200,000 cell/mL	299 (60)	885 (61)	1.58 (1.11–2.25)	0.010	1.39 (1.01–1.92)	0.045
>200,000 cells/mL	130 (27)	383 (26)	2.40 (1.67–3.46)	<0.001	1.92 (1.39–2.67)	<0.001
Region						
Hutterite colony farms	134 (27)	398 (27)				
North	16 (12)	46 (12)	Referent			
Central	37 (28)	110 (28)	1.70 (1.01–2.86)	0.047		
South	81 (60)	242 (61)	1.60 (0.98–2.61)	0.062		
Non-Hutterite farms	361 (73)	1,056 (73)				
North	123 (34)	357 (34)	Referent			
Central	159 (44)	467 (44)	0.78 (0.62–0.98)	0.036		
South	79 (22)	232 (22)	0.77 (0.57–1.05)	0.095		
Hutterite and non-Hutterite	495 (100)	1,454 (100)				
North	139 (28)	403 (28)			Referent	
Central	196 (40)	577 (40)			0.90 (0.75–1.08)	0.247
South	160 (32)	474 (33)			0.92 (0.75–1.14)	0.462
Farm type						
North region	139 (28)	403 (28)				
Non-colony farms	123 (88)	357 (89)	Referent			
Hutterite colony farms	16 (12)	46 (11)	0.61 (0.38–0.99)	0.046		
Central region	196 (40)	577 (40)				
Non-colony farms	159 (81)	467 (81)	Referent			
Hutterite colony farms	37 (19)	110 (19)	1.33 (0.97–1.82)	0.073		
South region	160 (32)	474 (33)				
Non-colony farms	79 (27)	232 (49)	Referent			
Hutterite colony farms	81 (73)	242 (51)	1.26 (0.92–1.73)	0.146		
All regions combined	495 (100)	1,454 (100)				
Non-colony farms	134 (27)	1,056 (73)			Referent	
Hutterite colony farms	361 (73)	398 (27)			1.23 (1.03–1.47)	0.020

¹Estimates are derived from 2 modified Poisson regression models: (1) a mixed-effects model accounting for repeated measures across all samples ($n = 1,454$) from 495 herds and (2) a standard model evaluating herd-level positivity based on at least 1 positive result among 3 samples.

²Herd size is categorized based on delivered milk volumes with $\leq 3,600$ L/d as small, 3,600–7,200 L/d as medium, and $> 7,200$ L/d as large.

³Approximate adult cows number estimated to ease interpretation of size categories using annual milk delivered by each farm and Alberta Milk's official statistics on average herd size and total Alberta milk production. This corresponds to approximately ≤ 125 adult cows for small, 125–250 adult cows for medium, and > 250 adult cows for large herd size (Shaukat et al., 2024).

⁴Annual production-weighted geometric mean bulk tank SCC (November 16, 2021–November 15, 2022).

Table 6. Production-weighted geometric mean bulk tank SCC (PWgBTSCC, $\times 1,000$ cells/mL) for Alberta dairy herds from November 16, 2021, to November 15, 2022, stratified by *Staphylococcus aureus* status and herd characteristics (region, herd size, and farm type)

Item	PWgBTSCC ($\times 1,000$ cells/mL): mean (95% CI); median (Q1–Q3)				
	All herds (n = 495)	<i>Staph. aureus</i> negative consistently (n = 212)	<i>Staph. aureus</i> positive at least once (n = 283)	<i>Staph. aureus</i> positive 1 time (n = 178)	<i>Staph. aureus</i> positive >1 time (n = 105)
Total Alberta herds	163 (158–169) 154 (118–202)	148 (140–156) 142 (109–178)	175 (167–182) 171 (128–214)	165 (156–174) 154 (123–205)	191 (179–203) 189 (144–229)
Region					
North Alberta	173 (162–184) 163 (117–220) ^a	151 (135–166) 145 (105–186) ^{a,i}	188 (173–203) 192 (127–230) ^{a,ii}	177 (159–196) 177 (126–220) ^a	202 (177–227) 208 (142–247) ^a
Central Alberta	170 (161–179) 168 (123–210) ^a	153 (140–166) 147 (111–187) ^{a,i}	184 (172–195) 183 (137–219) ^{a,ii}	176 (161–191) 176 (130–211) ^a	196 (178–214) 198 (170–231) ^a
South Alberta	146 (138–154) 143 (112–172) ^b	139 (127–151) 134 (110–165) ^{a,i}	151 (141–162) 145 (122–176) ^{b,i}	144 (132–156) 142 (105–172) ^b	168 (150–186) 156 (141–191) ^a
Herd size ^{1,2}					
Small	163 (155–171) 149 (118–204) ^a	152 (141–163) 144 (114–178) ^{a,i}	172 (161–184) 161 (123–216) ^{a,ii}	161 (148–175) 146 (119–203) ^a	194 (172–215) 198 (136–236) ^a
Medium	156 (148–165) 152 (113–190) ^a	131 (120–143) 121 (97–161) ^{b,i}	172 (161–183) 165 (134–200) ^{a,ii}	164 (149–179) 156 (127–189) ^a	183 (166–200) 177 (146–207) ^a
Large	180 (167–192) 189 (142–226) ^b	168 (148–187) 163 (129–202) ^{a,i}	188 (172–204) 201 (148–228) ^{a,i}	179 (158–200) 199 (125–226) ^a	202 (176–229) 211 (175–231) ^a
Farm type					
Hutterite colony	148 (139–157) 142 (110–180) ^a	134 (120–149) 130 (93–166) ^{a,i}	157 (145–168) 145 (122–192) ^{a,ii}	148 (134–161) 140 (110–175) ^a	178 (158–197) 175 (143–206) ^a
Non-colony	169 (162–175) 160 (121–210) ^a	152 (143–161) 147 (110–182) ^{a,i}	182 (173–191) 178 (135–222) ^{b,ii}	174 (163–185) 174 (126–213) ^b	195 (180–209) 191 (147–231) ^a
Mean PWgSCC ²					
<100,000	79 (75–82) 81 (69–90)	77 (73–81) 80 (66–89)	81 (75–87) 87 (74–92)	81 (74–88) 87 (75–92)	82 (63–101) 82 (73–90)
100,000–200,000	147 (143–150) 147 (124–170)	144 (139–149) 144 (120–163)	149 (144–153) 147 (126–174)	147 (142–153) 145 (125–173)	152 (144–159) 151 (128–175)
>200,000	244 (237–251) 230 (214–261)	243 (229–257) 227 (215–261)	244 (236–252) 231 (214–264)	242 (231–252) 228 (213–269)	247 (234–259) 232 (217–257)

^{a,b}Within a column, medians of different categories of the same variable without a common superscript differ ($P < 0.05$) by Wilcoxon rank-sum test (for 2 groups) or by Kruskal–Wallis test (for 3 or more groups) with post hoc pairwise comparisons by Dunn's test and Bonferroni correction.

^{i,ii}Within a row, medians of consistently negative herds and herds positive at least once without a common superscript differ ($P < 0.05$) by Wilcoxon rank-sum test.

¹Herd size is categorized based on delivered milk volumes with $\leq 3,600$ L/d milk delivered as small, 3,600–7,200 L/d milk delivered as medium, and $> 7,200$ L/d milk delivered as large.

²Approximate adult cows number estimated to ease interpretation of size categories using annual milk delivered by each farm and Alberta Milk's official statistics on average herd size and total Alberta milk production. This corresponds to approximately ≤ 125 adult cows for small, 125–250 adult cows for medium, and > 250 adult cows for large herd size (Shaukat et al., 2024).

109,000–178,000]) compared with herds that tested positive at least once (171,000 cells/mL [128,000–214,000]). Regional differences were observed, with herds in southern Alberta having lower PWgBTSCC (143,000 cells/mL [112,000–172,000]) than those in central (168,000 cells/mL [123,000–210,000]) and northern (163,000 cells/mL [117,000–220,000]) regions (Figure 2). Larger herds had a higher PWgBTSCC compared with small and medium herds, whereas Hutterite colony herds had a lower count compared with non-colony herds. The PWgBTSCC was higher in *Staph. aureus*-positive herds across all subgroups, except for large herds and herds in the southern region, where no significant difference was observed (Table 6). A total of 45 Alberta herds were reclassified into lower categories when their annual PWgBTSCC is used rather than PWaBTSCC (Table 7).

The multivariable linear regression model fitted to log-transformed annual PWgBTSCC data revealed that herds testing positive at least once for any of the 3 mastitis pathogens had higher annual log-transformed PWgBTSCC compared with consistently negative herds ($\beta = 0.172$; 95% CI = 0.104, 0.239; $P < 0.001$), corresponding to an estimated 18.8% increase on the original scale (Table 8). After adjusting for mastitis status, herds in southern Alberta had a lower PWgBTSCC than those in the north ($\beta = -0.119$ [–0.210, –0.028]), corresponding to an estimated 11.2% decrease, whereas PWgBTSCC was not different between central and northern herds ($P = 0.86$). Hutterite colony herds tended to have a 6.9% lower PWgBTSCC ($\beta = -0.072$ [–0.154, 0.010]) than non-colony herds ($P = 0.086$) after adjusting for mastitis status and other covariates (Table 8).

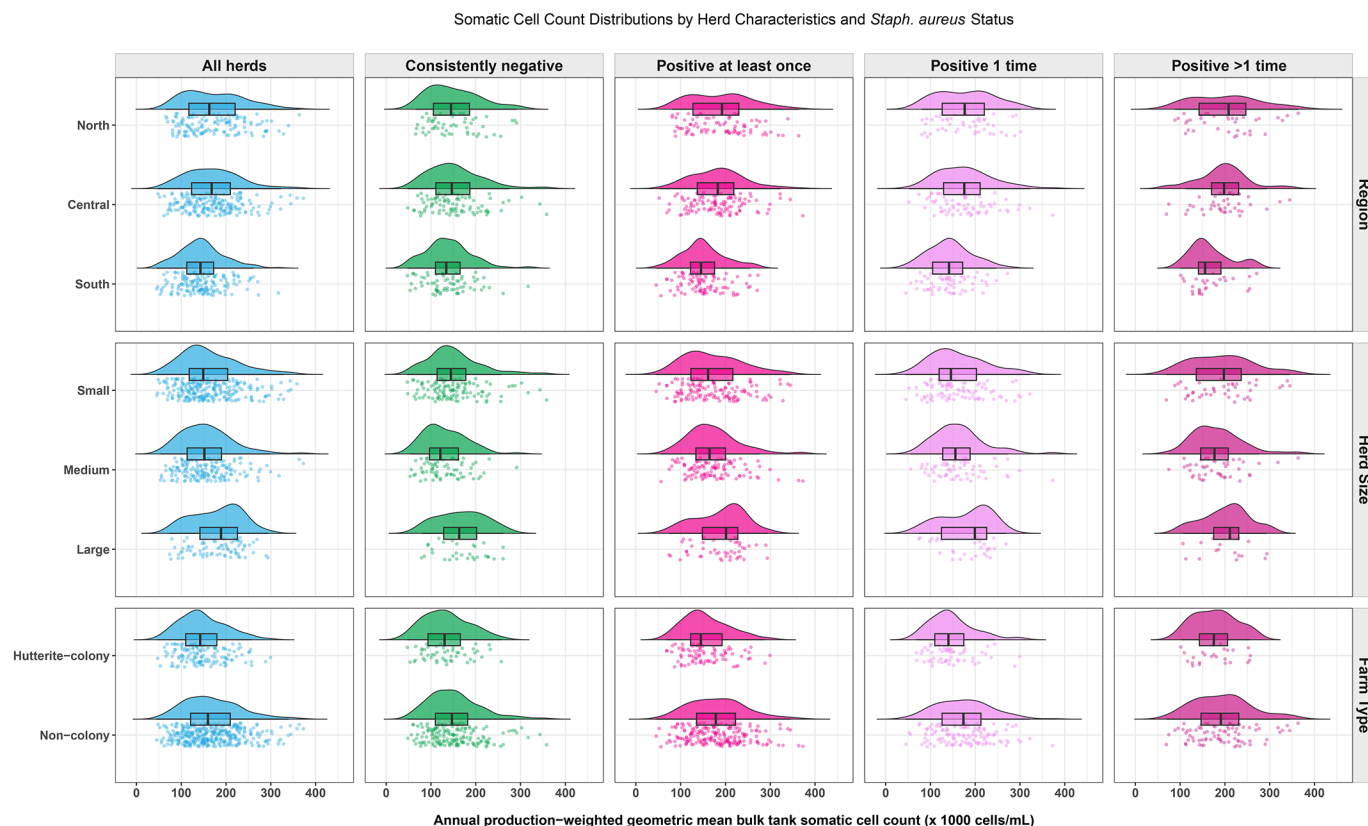


Figure 2. Distributions of annual production-weighted geometric mean bulk tank SCC (PWgBTSCC) in dairy herds, categorized by *Staphylococcus aureus* status and herd characteristics (geographical region, herd size, and herd type). Data are faceted by herd characteristics (rows) and *Staph. aureus* PCR results (columns). Each raincloud plot visualizes the density distribution (half-violin), the median and interquartile range (boxplot), and individual data points (jittered points).

Time-series plots of weekly PWgBTSCC over a 2-yr interval revealed that values for Alberta dairy herds ranged from 141,000 to 201,000 cells/mL, with higher levels during July to September (Figure 3). In 2021, PWgBTSCC ranged from 153,000 to 201,000 cells/mL, whereas in 2022, PWgBTSCC ranged from 141,000 to 198,000 cells/mL. From November 2021 to November 2022, herds consistently testing negative for *Staph. aureus* had PWgBTSCC values ranging from 126,500

to 178,500 cells/mL, whereas herds testing positive for *Staph. aureus* at least once had higher counts, ranging from 152,000 to 212,500 cells/mL, following similar seasonal patterns. Regional differences were also evident; herds in the southern region ranged from 125,500 to 185,500 cells/mL, whereas those in the northern region ranged from 156,500 to 221,000 cells/mL over the 2-yr interval. Non-Hutterite herds ranged from 145,000 to 208,000 cells/mL, and Hutterite colony herds ranged

Table 7. Comparing production-weighted geometric and arithmetic mean bulk tank SCC ($\times 1,000$ cells/mL) for Alberta dairy herds from November 16, 2021, to November 15, 2022

BTSCC category ($\times 1,000$ cells/mL)	Annual PW geometric BTSCC ($\times 1,000$ cells/mL)			Annual PW arithmetic BTSCC ($\times 1,000$ cells/mL)			No. of farms moved to a lower category ¹
	Mean (95% CI)	Median (Q1–Q3)	No. of farms	Mean (95% CI)	Median (Q1–Q3)	No. of farms	
<100	79 (75–82)	81 (69–90)	66	81 (78–85)	84 (73–93)	63	—
100–150	125 (123–128)	126 (113–138)	166	127 (125–129)	128 (115–139)	147	3
150–200	173 (171–176)	173 (159–186)	133	172 (170–175)	172 (159–184)	139	22
200–300	234 (229–239)	227 (213–255)	117	236 (231–240)	231 (215–253)	129	16
>300	333 (320–347)	329 (318–345)	13	339 (325–353)	339 (318–355)	17	4
Total	163 (158–169)	154 (118–202)	495	169 (164–175)	159 (124–210)	495	45

¹Number of farms that move to a lower BTSCC category if their annual geometric BTSCC is used rather than their arithmetic mean BTSCC.

Table 8. Coefficients and 95% CI from the multivariable linear regression model determining the association of natural log-transformed (log) annual production-weighted geometric mean bulk tank SCC (PWgBTSCC) with contagious mastitis pathogen status using PCR on bulk tank milk samples and herd characteristics using a multivariable linear regression model

Variable	Herds, no. (%)	Annual log (PWgBTSCC) ¹	
		Coefficient (95% CI)	P-value
Intercept		11.89 (11.81, 11.97)	
Contagious mastitis status			
Consistently negative herds	211 (43)	Referent	
Herds positive at least once for any of 3 mastitis pathogens	284 (57)	0.172 (0.104, 0.239)	<0.001
Herd size ^{2,3}			
Small	246 (50)	Referent	
Medium	172 (35)	−0.039 (−0.112, 0.035)	0.302
Large	77 (15)	0.083 (−0.017, 0.182)	0.103
Region			
North	139 (28)	Referent	
Central	196 (40)	−0.007 (−0.090, 0.075)	0.861
South	160 (32)	−0.119 (−0.210, −0.028)	0.011
Farm type			
Non-colony farms	134 (27)	Referent	
Hutterite colony farms	361 (73)	−0.072 (−0.154, 0.010)	0.086

¹Annual production-weighted geometric mean bulk tank SCC (November 16, 2021–November 15, 2022).

²Herd size is categorized based on delivered milk volumes with ≤3,600 L/d milk delivered as small, 3,600–7,200 L/d milk delivered as medium, and >7,200 L/d milk delivered as large.

³Approximate adult cows number estimated to ease interpretation of size categories using annual milk delivered by each farm and Alberta Milk's official statistics on average herd size and total Alberta milk production. This corresponds to approximately ≤125 adult cows for small, 125–250 adult cows for medium, and >250 adult cows for large herd size (Shaukat et al., 2024).

from 131,500 to 185,000 cells/mL in the same interval (Figure 3).

Association of Mastitis Positivity with BTSCC

Mixed-effect linear regression models revealed that BTM positivity for contagious mastitis pathogens was most strongly associated with elevated PWgBTSCC within a 7-d window centered on mastitis testing ($\beta = 0.067$ [95% CI = 0.032, 0.101]; $P < 0.001$; Table 9). However, as the time window was expanded, the strength of the association gradually declined, with β values of 0.059, 0.046, 0.037, and 0.031 for windows of 15, 30, 45, and 60 d, respectively. The effect was even milder at 75-d ($\beta = 0.024$; $P = 0.068$) and 90-d windows ($P = 0.085$). Across all time windows, PWgBTSCC was significantly lower during the April testing period compared with December, but higher during October compared with December. Furthermore, PWgBTSCC levels were lowest in spring (Table 9).

DISCUSSION

Here we describe the most current herd-level prevalence estimates for 3 of the most important contagious mastitis pathogens at 3 time points in a year in Alberta, Canada. As expected, the prevalence of herds positive

for *Strep. agalactiae* and *M. bovis* was low; however, *Staph. aureus* was the most prevalent contagious mastitis pathogen in Alberta, with some seasonal variation. Furthermore, important variations in prevalence of mastitis-positive herds across various herd characteristics and geographical regions were demonstrated. We also demonstrated distinct longitudinal BTSCC patterns in Alberta dairy herds, including seasonal fluctuations and differences by herd characteristics. Specifically, there were consistently higher BTSCC in herds positive for *Staph. aureus*, in non-Hutterite colonies, or in northern Alberta.

Our findings were consistent with other Canadian studies, concluding that the prevalence of herds positive for *M. bovis* was very low in Canada. Only 0.5% herds of the 372 herds tested across Canada were positive for *M. bovis* in Canadian NDS (Bauman et al., 2018). Others also reported a low prevalence for *M. bovis*-positive herds across various Canadian provinces, including 3% in Ontario (Nobrega et al., 2024), 3% in Québec (Francoz et al., 2012), and 2% in Prince Edward Island (Olde Riekerink et al., 2006). Although mastitis-associated *M. bovis*-positivity in Canadian dairy herds is generally low, *M. bovis* is increasingly becoming a pathogen of concern for Canadian dairy farmers. A preliminary report (Waller, 2024) recently indicated that 17% dairy farms in western Canada are likely harboring *M. bovis*; however, limited

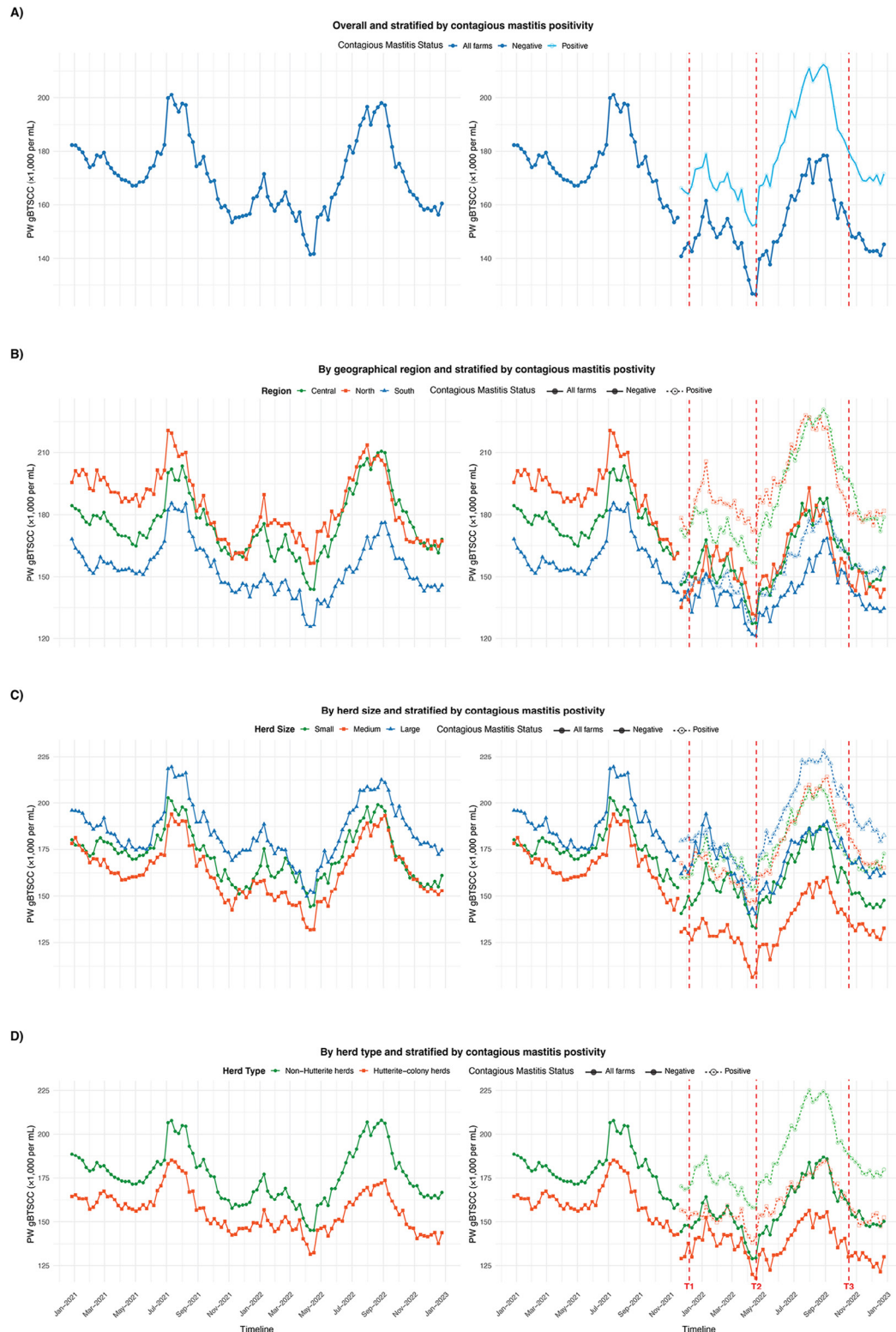


Figure 3. Weekly trends in production-weighted geometric mean bulk tank SCC (PWgBTSCC; $\times 1,000$ cells/mL) among dairy herds in Alberta from January 2021 to December 2022. Panel A shows the overall trend, whereas panels B to D display trends stratified by (B) region (north, central, and south), (C) herd size (small, medium, large), and (D) herd type (Hutterite colony vs. non-colony herds). Each panel contains 2 plots: the left plot shows overall PWgBTSCC by subgroup, and the right plot further stratifies PWgBTSCC by bulk tank PCR status for contagious mastitis pathogens (solid lines: PCR-negative; semi-transparent lines: PCR-positive). Red dotted vertical lines indicate the 3 time points when mastitis PCR testing was conducted.

Table 9. Adjusted coefficients and 95% CI from mixed-effect linear regression models for associations between natural log-transformed (log) production-weighted geometric mean bulk tank SCC (PWgBTSCC) and PCR positivity of bulk tank milk for *Staphylococcus aureus*, *Streptococcus agalactiae*, or *Mycoplasma bovis*¹

Outcome	Model output	PCR status for contagious mastitis			Time point		
		Negative	Positive		December 2021	April 2022	October 2022
Herd, no. (%)		211 (43)	284 (57)		484 (33)	486 (34)	484 (33)
Samples, no. (%)		1,048 (72)	406 (28)		484 (33)	486 (34)	484 (33)
log PWgBTSCC, 7 d	Coefficient (95% CI)	Referent	0.067 (0.032, 0.101)		Referent	-0.108 (-0.139, -0.078)	0.079 (0.049, 0.11)
	P-value		<0.001			<0.001	<0.001
log PWgBTSCC, 15 d	Coefficient (95% CI)	Referent	0.059 (0.028, 0.091)		Referent	-0.085 (-0.113, -0.058)	0.096 (0.068, 0.124)
	P-value		0.001			<0.001	<0.001
log PWgBTSCC, 30 d	Coefficient (95% CI)	Referent	0.046 (0.017, 0.076)		Referent	-0.069 (-0.095, -0.043)	0.084 (0.058, 0.11)
	P-value		0.002			<0.001	<0.001
log PWgBTSCC, 45 d	Coefficient (95% CI)		0.037 (0.009, 0.066)		Referent	-0.061 (-0.086, -0.036)	0.08 (0.055, 0.104)
	P-value		0.011			<0.001	<0.001
log PWgBTSCC, 60 d	Coefficient (95% CI)	Referent	0.031 (0.003, 0.058)		Referent	-0.049 (-0.073, -0.025)	0.078 (0.054, 0.102)
	P-value		0.029			<0.001	<0.001
log PWgBTSCC, 75 d	Coefficient (95% CI)		0.024 (-0.002, 0.051)			-0.048 (-0.07, -0.025)	0.077 (0.055, 0.100)
	P-value		0.068			<0.001	<0.001
log PWgBTSCC, 90 d	Coefficient (95% CI)	Referent	0.022 (-0.003, 0.048)		Referent	-0.043 (-0.065, -0.021)	0.079 (0.057, 0.101)
	P-value		0.085			<0.001	<0.001

¹Coefficients represent the estimated change in log PWgBTSCC.

information was provided about methodology and sampling strategies. In contrast, the low prevalence of *M. bovis*-positive herds estimated in our study and previous ones is likely related to a lower sensitivity of testing BTM for *M. bovis* with the goal of detecting nonactive infections caused by *M. bovis*. Although our TP estimates were adjusted for a lower sensitivity of qPCR, our latent herd-level infection state definition was a situation with both free *M. bovis* DNA as well as antibodies in BTM, which is related to the presence of a certain level of currently infected animals in the herd (Nielsen et al., 2015). Of note, *M. bovis* does not necessarily cause mastitis and may have other manifestations, including arthritis, otitis media, and respiratory diseases, especially in youngstock that are unlikely to be detected using BTM.

Herd-level prevalence of *Strep. agalactiae* has decreased in Canada and nearly all countries with a developed dairy industry as a result of the implementation of the 5-point contagious mastitis control program in the dairy industry (Neave et al., 1969). In Alberta, *Strep. agalactiae* prevalence decreased from 21% in 1986 to 11% in 1992 and 6% in 1993 (Schoonderwoerd et al., 1993). Our findings were in line with previous studies from various parts of Canada reporting rare positivity for *Strep. agalactiae*. The Canadian NDS reported only 0.3% herds positive for *Strep. agalactiae*, whereas other studies described 0.1% in Ontario (Nobrega et al., 2024), 1.6% in Prince Edward Island (Olde Riekerink et al., 2006), but much higher (7.7%) in Québec (Francoz et al., 2012). It should be noted that the last 2 studies are >10 yr old.

Prevalence of herds that were at least once PCR-positive for *Staph. aureus* in Alberta was high in the current study, consistent with earlier studies from Prince Edward Island (74%), Québec (85%), and Ontario (50%; Olde Riekerink et al., 2006; Francoz et al., 2012; Nobrega et al., 2024). Positivity rates at a single time point were lower than those reported in other Canadian provinces, in line with the NDS, which reported a higher apparent positivity of *Staph. aureus* in Atlantic Canadian provinces (40%–70%), Ontario (39%), and Québec (62%) than in western Canadian provinces such as Alberta (25%) and British Columbia (10%; Bauman et al., 2018). This likely reflects differences in management practices across these provinces. Regardless, *Staph. aureus* remains a major contagious mastitis pathogen in Canada.

The lower TP relative prevalence to AP estimates reflects the lower Sp (67%) of the qPCR method when applied to BTM samples in accurately identifying herds with a minimum within-herd prevalence of cows experiencing subclinical mastitis due to active *Staph. aureus* infection detectable through bacterial culture. Test characteristics used in our Bayesian latent class analysis were informed by a prior study that evaluated the same qPCR

kit (M4E) on BTM samples, comparing its performance against bacterial culture results from individual cows' quarter-level milk samples contributing to BTM (Soltau et al., 2017). Another study, using Bayesian latent class analysis in the absence of a gold standard, compared 3 tests (qPCR and 2 bacterial cultures [quantitative and qualitative]) applied to BTM samples, and similarly reported a lower specificity of 67% for qPCR (Zanardi et al., 2014). Consequently, the latent class infection status in our model was effectively defined as herds with active *Staph. aureus* IMI detectable via bacterial culture. An important consideration is that PCR detects DNA from both viable and nonviable bacteria, whereas culture only detects viable organisms. Consequently, qPCR-positive but culture-negative results are classified as false positives in our latent class framework, although such discordance may alternatively reflect culture insensitivity rather than qPCR false positives. This diagnostic complexity warrants careful interpretation of conflicting results. Additionally, whereas the analytical specificity of qPCR is high and PCR signals reliably indicate *Staph. aureus* DNA presence, contamination of BTM samples from non-udder sources remains an important consideration. For instance, *Staph. aureus* DNA detected in BTM may have originated from teat skin, teat canal infections, or environmental contamination rather than true IMI (Haveri et al., 2008; Capurro et al., 2010). A study from Ontario also estimated a lower TP estimate compared with PCR positivity for *Staph. aureus* (31% vs. 51%) using BTM (Nobrega et al., 2024). Our TP estimates were higher when a higher Sp was assumed in our analysis, indicating that assumptions about test Sp substantially influence TP estimates, highlighting the uncertainty inherent in diagnostic-based surveillance and the need for robust validation of test characteristics. This highlights the need for cautious interpretation of qPCR-positive, culture-negative results and supports the importance of integrating molecular and bacteriological diagnostics for more reliable herd-level surveillance.

Importantly, TP estimates discussed likely represent the proportion of Alberta dairy herds in which *Staph. aureus*-associated IMI was present at a single moment. However, this estimate is subject to an important limitation: Se of qPCR using BTM will vary according to the definition of a positive herd, which can be based on within-herd prevalence levels (Soltau et al., 2017). Specifically, BTM PCR requires ~27.6% of cows to be infected to achieve 90% probability of detection, or 15.5% of cows for 70% probability of detection. Although cow-level prevalence data in Canada are scarce, a study based on the National Cohort of Dairy Farms (Reyher et al., 2011) from various Canadian regions reported a median herd quarter-level *Staph. aureus* prevalence of just 3.9% (Dufour et al., 2012). This suggests it is plausible that many herds in our

population have infection levels far below the within-herd threshold to achieve higher detection probability with BTM qPCR. Consequently, using a fixed sensitivity in our model may have led to an underestimation of the true herd-level prevalence. Our sensitivity analyses support this possibility, as assuming lower test sensitivities resulted in higher TP estimates. Establishing the distribution of within-herd *Staph. aureus* prevalence in Alberta is therefore a critical next step for validating and improving BTM-based surveillance programs.

Herds with PWgBTSCC >200,000 cells/mL had 2.4 times higher prevalence of *Staph. aureus*-positive herds compared with <100,000 cells/mL herds. These findings were similar to a previous Canadian study reporting that farms with SCC levels >200,000 had 1.5 times higher odds of a *Staph. aureus*-positive BTM compared with herds with <200,000 (Bauman et al., 2018). Seasonal fluctuations were observed in BTSCC levels, with peaks observed during late summers, similar to previous studies reporting higher BTSCC levels and mastitis incidence during summer months in Canada and elsewhere (Olde Riekerink et al., 2007; Shock et al., 2015).

The higher PWgBTSCC observed in larger herds was evidently driven by contagious mastitis pathogens, as the association disappeared after adjusting for mastitis positivity. Interestingly, medium-sized herds were more frequently PCR-positive compared with smaller herds, whereas large herds did not differ significantly. This may be attributed to the dilution effect: If large herds maintain a lower within-herd prevalence, the greater milk volume from healthy cows can dilute pathogen DNA below the PCR detection threshold. This is also supported by longitudinal patterns we observed, where differences between PCR-positive and negative herds were much stronger in medium- versus large-size herds, suggesting that PCR-negative larger herds may still have higher SCC cows.

Herds in northern Alberta were more frequently tested positive for *Staph. aureus* compared with those in the central or southern regions. Although data on herd-level management practices were not available in this study, such regional disparities may reflect differences in milking hygiene, housing systems, bedding material, outdoor access, or environmental conditions known to influence mastitis incidence (Wenz et al., 2007; Olde Riekerink et al., 2010). Interestingly, herds in the south region consistently had lower BTSCC levels, even after stratifying by contagious mastitis PCR-positivity status. A similar trend was also evident among Hutterite colony herds, which maintained lower levels of BTSCC despite higher *Staph. aureus* positivity overall. However, Hutterite colony status did not appear to be a significant predictor of BTSCC in multivariable analysis, adjusting for mastitis status. Given that approximately half of all dairy herds in the southern region are Hutterite colonies, this may partly

explain the region's overall lower BTSCC, although non-Hutterite herds in this region also had lower levels. Taken together, these patterns suggest that region and herd type likely act as markers for unmeasured management differences, rather than being causal determinants themselves. Therefore, the regional and herd-type associations reported here should be viewed as population-level signals describing risk stratification across strata of herd characteristics that can guide target surveillance and hypothesis generation, rather than as directly modifiable factors for herd-level mastitis control.

However, *Staph. aureus* positivity among Hutterite herds varied by region; southern colony herds were more frequently positive than northern ones, whereas the opposite pattern was observed among non-Hutterite herds. Such regional and herd-type disparities highlighted the need for future studies to account for management practices, housing systems, and hygiene protocols when comparing mastitis dynamics across regions. Therefore, future research should explicitly consider herd type (Hutterite vs. non-Hutterite) and geographical regions during farm enrollment to ensure valid comparisons and minimize selection bias.

This study was based on repeated BTM sampling from all active dairy herds in Alberta at 3 distinct time points. Bulk tank milk testing is increasingly recognized as a practical, noninvasive, and cost-effective approach for herd-level surveillance of infectious diseases, including contagious mastitis (Nobrega et al., 2023). However, this method carries inherent limitations. Because BTM represents pooled milk only from cows milked on the day of collection, it excludes animals with clinical mastitis or those under antimicrobial treatment, as their milk is typically withheld. Consequently, BTM results primarily reflect subclinical infections rather than the full spectrum of mastitis within a herd. In addition, imperfect Se and Sp of qPCR mean that some herds were misclassified for *Staph. aureus* status. Therefore, the associations we report between herd-level characteristics and BTM qPCR positivity should be interpreted as relationships with a surveillance measure, and some of these associations are likely weaker than they would be if true herd infection status could be observed directly.

Moreover, the milking system (automatic milking system [AMS] vs. conventional) was not evaluated in this study in relation to qPCR positivity or BTSCC trends. Although approximately 29% of Alberta dairy herds used AMS during the study period, information on AMS type and model, timing of conversion, and other time-varying herd changes around adoption was not available, limiting the interpretation of any AMS comparison in this dataset. Future studies with more detailed milking systems and transition information could evaluate associations with mastitis pathogen status and BTSCC more directly.

Our study was confined to Alberta dairy herds, yet this province is also a major hub for beef production. Given the geographical proximity of intensive dairy and beef production in Alberta, and shared use of transportation networks, markets, and some service providers, pathogens such as *M. bovis*, a known agent of both mastitis and bovine respiratory disease, may circulate between dairy and beef operations. Previous reports of *Mycoplasma* prevalence in western Canadian bison herds further suggest possible cross-species or interherd transmission routes (Bras et al., 2016, 2017). Future research should therefore explore potential epidemiological links among dairy, beef, and wildlife populations to better understand the ecology of *Mycoplasma* in the region.

Overall, true prevalence estimates and BTSCC dynamics presented here provide insights for guiding future research, extension activities, and policy development aimed at improving udder health and milk quality in western Canada and beyond. Our surveillance framework, using multiplex qPCR on routinely collected BTM samples and Bayesian latent class modeling, enables pathogen-specific true herd-level prevalence estimation at multiple time points while accounting for imperfect test performance. When integrated with herd metadata and routine BTSCC recording, this approach can generate stratified baseline estimates, track temporal changes, and identify higher-risk herd strata for targeted follow-up within-herd testing. Because BTM sampling for quality testing and BTSCC recording are part of many milk quality programs, this framework can be adapted to regions with centralized milk collection and tailored to local management practices and pathogen profiles. Our analytic approach applies to cross-sectional prevalence studies with common outcomes, where estimating PR instead of OR can reduce the risk of misinterpretation. In addition, adopting a serial cross-sectional design rather than a single cross-section to address variability in bulk tank composition over time provides a more informative framework for enhanced, targeted surveillance.

CONCLUSIONS

More than half of dairy herds in Alberta tested PCR-positive for *Staph. aureus* using BTM at least once across 3 sampling events in a year, although the estimated true herd-level prevalence was lower and varied across the 3 time points. Among the 3 contagious pathogens, the true herd-level prevalence was highest for *Staph. aureus*, followed by *M. bovis*, with *Strep. agalactiae* detected in very few herds. Herds located in the northern region of Alberta were more frequently *Staph. aureus* positive. Additionally, Hutterite colony herds had a higher positivity for *Staph. aureus* at least once. Bulk tank SCC was lower in herds from the south region. Moreover, BTSCC

was elevated during a 60-d window surrounding mastitis testing but was no longer different beyond this period. Beyond Alberta, applying this surveillance and analytical framework to routine bulk tank and herd-level data in other dairy regions enables current, pathogen-specific herd-level prevalence estimates and supports benchmarking, risk stratification, early detection of population-level shifts, and targeted deployment of surveillance and control resources.

NOTES

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Nonstandard abbreviations used: AMS = automatic milking system; AP = apparent prevalence; BCI = Bayesian credible interval; BTM = bulk tank milk; BTSCC = bulk tank somatic cell count; ESS = effective sample sizes; M4E = Mastitis 4E; NDS = National Dairy Study; OR = odds ratio; PR = prevalence ratio; PWgBTSCC = production-weighted geometric mean BTSCC; Q1 = first quartile; Q3 = third quartile; qPCR = quantitative PCR; Se = sensitivity; Sp = specificity; T1 = December 2021; T2 = April 2022; T3 = October 2022; TP = true prevalence; VIF = variance inflation factors.

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ORCID

Waseem Shaukat, <https://orcid.org/0000-0002-9117-3438>
 Diego B. Nobrega, <https://orcid.org/0000-0001-9821-1436>
 Kristen N. Kalbfleisch, <https://orcid.org/0009-0005-4482-8461>
 John P. Kastelic, <https://orcid.org/0000-0003-4607-3355>
 David F. Kelton, <https://orcid.org/0000-0001-9606-7602>
 Herman W. Barkema <https://orcid.org/0000-0002-9678-8378>